Evaluation

- ☑ Multiplex polymerase chain reaction (PCR) studies (FilmArray panel) can be performed whenever feasible and especially if antimicrobials are given already.
- ☑ The various CSF findings are given in **Table 3**.

TABLE 3: Cerebrospinal fluid (CSF) findings.

Investigation	*Normal*	*Bacterial*	*Viral*	*Tuberculosis*	*Fungal*
Opening pressure	10–20 cm	High	Normal/high	High	High/very high
Color	Clear	Cloudy	"Gin" color	Cloudy/yellow	Clear/cloudy
Cells	<5	High/very high	Slightly increased	Slightly increased	Normal-high
		100–50,000	5–1,000	<500	0–1,000
Differential	Lymphocytes	Neutrophils	Lymphocytes	Lymphocytes	Lymphocytes
CSF/plasma glucose	50–66%	Low <40%	Normal	Low-very low (<30%)	Normal-low
Protein (g/L)	<0.45	High	Normal-high	High-very high	Normal-high
		>1	0.5–1	1.0–5.0	0.2–5.0

Management

- ☑ The main aim is for initial stabilization, specific diagnosis, and preventing further brain injury, symptomatic treatment, and specific therapies.
- ☑ All patients with Glasgow Coma Scale (GCS) < 8 should be managed in intensive care unit (ICU). Airway should be secured by intubation and proper ventilation should be maintained. Circulation should be secured by an IV line after taking necessary samples and fluid boluses or inotropes should be initiated in cases of shock.
- ☑ Supportive care includes maintaining euvolemia, euglycemia, euthermia, and avoiding dyselectrolytemia.
- ☑ Raised intracranial pressure (ICP) is managed with neuroprotective measures and osmolar therapy (mannitol, hypertonic saline, and hyperventilation) while keeping in mild head-end elevation—15–30°.
- ☑ Seizures are managed as per standard guidelines of treatment of acute symptomatic seizures (first-line—benzodiazepines, second-line—phenytoin/levetiracetam/phenobarbitone/sodium valproate, all being equally effective).

Management

- ☑ Empirical treatment with third-generation cephalosporins in suspected bacterial etiology—ceftriaxone or cefotaxime.
- ☑ In sporadic viral encephalitis, acyclovir should be started at 20 mg/kg/dose, 8th hourly 14–21 days depending on the response and investigation results.
- ☑ Further addition/titration of antimicrobials is done on basis of investigations and PCR studies.
- ☑ Immunotherapy in the form of steroids and intravenous immunoglobulin IVIG, if autoimmune encephalitis is suspected.
- ☑ For treatment of complications and long-term management, if sequelae—physiotherapy, prevention of bed sores and exposure keratitis, proper nutrition [either orally or through nasogastric (NG) feeds], and psychological support to patient and family.

Further Reading

- ☑ Gupta P. PG Textbook of Pediatrics, 2nd edition. New Delhi: Jaypee Brothers Medical Publishers (P) Ltd.; 2017.
- ☑ Sharma S, Mishra D, Aneja S, Kumar R, Jain A, Vashishtha VM. Consensus guidelines on evaluation and management of suspected acute viral encephalitis in children in India. Indian Pediatr. 2012;49:897-910.

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Phimosis

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